

Lesson 8-1: Understand Equations and Their Solutions

Grade 6 Mathematics · Standard 6.AT.8

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINT

Which equation represents: 'A number plus 12 equals 30'?

- ☐ A $n + 12 = 30$
- ☐ B $n - 12 = 30$
- ☐ C $12n = 30$
- ☐ D $n / 12 = 30$

Question 2 SELECTED RESPONSE · 1 POINT

Which equation represents: 'Seven less than a number is 18'?

- ☐ A $n - 7 = 18$
- ☐ B $7 - n = 18$
- ☐ C $n + 7 = 18$
- ☐ D $n / 7 = 18$

Question 3 SELECTED RESPONSE · 1 POINT

Which equation represents: 'Three times a number equals 21'?

- ☐ A $3n = 21$
- ☐ B $n + 3 = 21$
- ☐ C $n - 3 = 21$
- ☐ D $n / 3 = 21$

Question 4 SELECTED RESPONSE · 1 POINT

Is $n = 3$ a solution of $n + 5 = 9$?

- ☐ A Yes, because 3 is smaller than 9
- ☐ B No, because $3 + 5 = 8$, which does not equal 9
- ☐ C Yes, because $3 + 5$ is close to 9
- ☐ D You cannot tell without solving first

Question 5 SELECTED RESPONSE · 1 POINT

Which equation represents: 'A number divided by 4 equals 8'?

- ☐ A $n / 4 = 8$
- ☐ B $4n = 8$
- ☐ C $n - 4 = 8$
- ☐ D $n + 4 = 8$

Question 6 SELECTED RESPONSE · 1 POINT

A detective found some fingerprints on Monday and 14 more on Tuesday, for a total of 30. Which equation represents this?

- ☐ A $f + 14 = 30$
- ☐ B $f - 14 = 30$
- ☐ C $14f = 30$
- ☐ D $f / 14 = 30$

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

Marisol tests values in the equation $7 + m = 23$. Which value of m makes the equation true?

Fill in the bubble next to the one best answer.

- ☐ A $m = 9$
- ☐ B $m = 16$
- ☐ C $m = 23$
- ☐ D $m = 30$

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Which reasoning best justifies that answer?

- ☐ A Adding the two numbers written in the equation, $7 + 23$, gives the value of m .
- ☐ B The equal sign means the answer is always the number written to the right of it.
- ☐ C Substituting 16 for m gives $7 + 16 = 23$, so both sides of the equation have the same value.
- ☐ D Any number can replace m because m is a variable, and variables can change.

Question 8**SELECT ALL THAT APPLY**

Select ALL statements that are true about equations and their solutions.

Mark the box next to EVERY answer that is true. More than one is correct.

- ☐ A An equation is a math sentence that uses an equal sign to show two expressions have the same value.
- ☐ B You can test a value by substituting it for the variable and checking whether both sides are equal.
- ☐ C The value $n = 6$ is a solution of $n + 9 = 15$ because $6 + 9 = 15$.
- ☐ D The expression $4y + 2$ is an equation because it contains a variable.
- ☐ E The value $x = 15$ is a solution of $x - 4 = 9$ because $15 - 4 = 9$.

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9**WRITTEN RESPONSE · 2 POINTS**

Write a real-world situation that can be modeled by the equation $n / 5 = 9$. Explain what n represents and verify your equation makes sense.

Write your answer in complete sentences. Show or explain your mathematical thinking.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.